

Draft thesis outline

Research Question

What is the significance of the recent proof of the point convergence of Nesterov's accelerated gradient method [1] in terms of the adjacent literature on acceleration of first-order methods for convex optimisation?

Goal of the thesis is to give an overview of the state of knowledge on this question, using [1] as an entry point and tracing a narrow path through the literature to this work starting from Nesterov's original proposal of his acceleration method [2].

Plan

1. Minimisation of smooth, convex functions

- Presentation of the problem of interest: $\min f$ where f is convex and L -smooth.
- Cover relevant important properties of convex and L -smooth functions. Why are these important or natural constraints on the problem? Give examples.
- Plain Gradient Descent method, give streamlined proofs of convergence in value and point convergence using Lyapunov function approach (e.g. from [3] but motivate the introduction of 'potential functions' more clearly).

2. Nesterov's accelerated gradient method and complexity

- Introduce concepts: algorithms as 'black boxes' that output sequences of given length; first-order methods as a class of algorithms; oracles and computational/informational cost.
- Nesterov's Accelerated Gradient Method, introduction, simplified proof given by Nesterov (use Lyapunov approach again [3]).
- Nimerovskii complexity result on the lower bound speed of convergence in value - Nesterov's method attains this value which is why it is interesting.
- Potentially mention other more recent complexity results, like [4].

3. Point convergence of accelerated methods

- Present the point convergence of Nesterov's gradient method from [1]
- Discuss FISTA method, point convergence, differences with NAG that allowed researchers to find the proof first (also discuss section 3 of Nesterov's original paper in this context).
- Present OGM in the context of PEP

4. Discuss potential paths forward

- Subject to further planning

Bibliography

- [1] U. Jang and E. K. Ryu, “Point Convergence of Nesterov's Accelerated Gradient Method: An AI-Assisted Proof.” 2026.
- [2] Y. E. Nesterov, “A method of solving a convex programming problem with convergence rate $O\left(\frac{1}{k^2}\right)$ ”, *Dokl. Akad. Nauk SSSR*, vol. 269, no. 3, pp. 543–547, 1983.
- [3] N. Bansal and A. Gupta, *Theory of Computing*, vol. 15, no. 1, pp. 1–32, 2019, doi: [10.4086/toc.2019.v015a004](https://doi.org/10.4086/toc.2019.v015a004).
- [4] Y. Drori, “The exact information-based complexity of smooth convex minimization,” *Journal of Complexity*, vol. 39, pp. 1–16, 2017, doi: [10.1016/j.jco.2016.11.001](https://doi.org/10.1016/j.jco.2016.11.001).